

WellDone! — Manual

Plate setup for the Roche LightCycler PRO

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1. What is WellDone!?

WellDone! is used for PCR setup on the LightCycler PRO. The app

- manages an **assay library** (reagents, controls, PCR profile, color),
- records your **samples** (manually, via barcode scanner or XML/CSV import from the Roche MagNA Pure 96),
- automatically computes the **plate setup** of the PCR plate,
- creates the **pipetting scheme (PDF)** and the **plate-setup file (CSV)** for import into the LightCycler PRO software,
- and remembers everything in a **session file** so you can reopen it later.

Scope: WellDone! supports the plate setup of **qualitative** tests (e.g. LightMix Modular from TIB Molbiol) as well as **quantitative** tests with an **external** standard curve – i.e. reference/melting standards (RS/MS) whose standard curve is stored/imported **on the instrument**. **In-run standard curves or dilution series** in the plate setup are **not** supported.

It works **completely offline** – no data is sent to the internet.

2. Important note (liability)

WellDone! is a helper tool, not a medical device, and is not validated. It is your responsibility to **check** the calculated plate setup, the control positions and the MasterMix volumes **before every run**. Do not transfer any values into diagnostic or otherwise critical workflows without checking them. **No liability** is accepted for damage, incorrect results or consequential costs. When in doubt, your established (validated) method applies.

3. Installation & first start

- **Operating system:** Windows 11 (Windows 10 should also work).
- **Installation:** none. WellDone! is a single file `WellDone.exe` – just double-click to start it, **no administrator rights** required.
- Put the `.exe` in a **fixed folder** (not in "Downloads"), e.g. `C:\WellDone\` – there you will find it again and can also place updates.
- Your **assay library** (`assays.db`) is automatically stored under `%APPDATA%\WellDone` – **regardless of where the `.exe` is located** – and is preserved when you replace the `.exe`.
- On the very first start, three example assays are present, which you can adapt or delete.

First start — SmartScreen note

On the **very first start**, Windows may show the blue notice **"Windows protected your PC"** (Microsoft Defender SmartScreen). This is **normal and not a virus** – it appears only because the app is new and not (yet) digitally signed. This is how you start it (one time only):

1. Click **"More info"**.
2. Then click **"Run anyway"**.

After that, Windows remembers the file – the notice will **not appear again** for this `.exe`.

Note for IT: WellDone! is a standalone, **unsigned** `.exe` – **no installation, no administrator rights, no system changes**, fully offline. The SmartScreen message is based only on missing "reputation"/code signature, **not** on malicious code. Under restrictive policies, IT can whitelist the file or its hash.

4. Layout of the window

The window is divided into **three columns** and **two rows**:

	Center: Plates	Right: Tools
Top left: Plate setup (session name, plate ID, comment, workflow checklist, <i>New · Load · Save</i>)	Sample plate (96 wells for your samples)	Sample-plate tools (input direction, source plate ID/label, field detail of the sample, <i>PDF · Import · MP96 Order · Undo · Clear samples</i>)
Bottom left: Assays (96/384 tabs, library with selection checkboxes, order, <i>New · Delete · Edit</i>)	PCR plate (96 or 384 wells, filled automatically)	PCR tools (<i>Run plate setup</i> , start-well hint, <i>Undo · Clear PCR</i> , layout toggle, field detail, assays overview, <i>Print · PDF · Setup PRO</i>)

5. The workflow in 7 steps

The **workflow checklist** at the top left ticks off these steps automatically (green):

1. **Enter / import samples** (sample plate)
2. **Select assays** (checkbox in the assay list)

3. **Run plate setup** (large blue button)
4. **Pipetting scheme printed / saved** (PDF)
5. **Plate setup exported** (CSV for LightCycler PRO)
6. **Save plate setup / session** (.wds)

7. **WellDone!** — appears in green once all steps are done.

If something changes (a new sample, a different assay ...), the "saved" ticks (4–6) are reset so you don't export anything outdated.

6. Managing assays

An **assay** describes a PCR setup. You maintain the library via *New*, *Edit*, *Delete* (at the bottom of the assay column). A **double-click** on an assay opens *Edit* directly. With **"Duplicate"** (on the right in the order row) you duplicate the selected assay (all fields; name "... (copy)").

Fields of an assay

Field	Meaning
Format (96/384)	Plate format of the assay. Set via the tab in the assay list; the LCAPs for 96 and 384 differ, so every assay belongs to exactly one format
Display name	Name in the app (max. 24 characters, so the PCR profile stays visible)
LCAP name	Name written into the plate-setup CSV as AssayName
PCR profile	Run protocol (RunProfileName in the CSV)
Controls	PTC, PPC, NPC, NTC – selectable as needed
Control placement	Where the controls sit on the plate (3 modes, see below)
Water / primer-probes / MasterMix / RT enzyme (µl)	Components per reaction
Template (µl)	Sample volume per well
Safety volume	Reserve against pipetting loss (see section 9). Default 1.0 ; adjustable 0 to 10 in steps of 0.1 (e.g. 0.5)
Standards (optional)	Reference/melting standards (RS/MS) per target – only for abs. quantification / MCGT with an external standard curve (see below)
Color	Color of the assay on the PCR plate

Control placement (selectable per assay in the Edit dialog) – how the controls sit on the plate:

- **Positive -> samples -> negative** (default): positive controls at the top (**PTC, PPC**), then the samples, the negative controls at the end (**NPC, NTC**).
- **Negative -> samples -> positive**: the same arrangement, just reversed (negatives at the top, positives at the end).
- **Samples -> positive -> negative**: samples first, **all controls at the end** (positives before negatives). Particularly for **quadrant stamping** on 384 (see section 8) – the samples stay in their position, the controls come afterwards.
- **Samples -> negative -> positive**: as before, just negatives before positives.
- **Fixed positions**: each active control gets a **fixed well** that you **enter as text** (e.g. PTC = A1, NTC = H12; on 384 up to P24). The **samples** flow normally and **skip** the occupied fixed wells. If several assays are active at the same time, their fixed wells must **differ**.

Analysis types & standards. The analysis type (qualitative, Tm calling, abs. quantification, MCGT) is in the **LCAP** – the plate-setup CSV has the same columns for all types. **Qualitative and Tm calling** need no standards (only samples + controls). For **abs. quantification** and **MCGT** you work with an **external standard curve**: the curve is **imported as a file** at the LightCycler PRO (it is **not** on the plate). On the plate you only **optionally** carry one standard per target:

- **RS – reference standard** (quantification, for calibration) or **MS – melting standard** (MCGT).
- In the Edit dialog under "**Standards**": add a row with *role (RS/MS) · name · target · note*. **Name** = standard/genotype name **exactly as in the LCAP**; **Target** = target nucleic acid (mandatory). Each standard occupies **one well** (like a control, at the end of the layout) and appears in the CSV with SampleRole RS/MS and the Targetname.
- **Note (optional)**: free documentation, e.g. concentration/titer or tube ID. It appears in the **pipetting-scheme PDF** (legend "Standards") and in the tooltip, **not** in the CSV. The actual **concentration/titer is maintained in the LCAP** (on the instrument), not in the plate setup.

Rule – same PCR profile: only assays that use the **same PCR profile** may be selected at the same time. As soon as one assay is active, incompatible assays are greyed out; when you try to tick an incompatible one, a note appears.

Tabs 96 / 384: you choose the format at the top of the assay list. Only assays of the format that matches the current **PCR plate size** can be ticked; you can open the other tab at any time to **view/edit**. More on 384 in section 8.

Order of the layout: with the arrows ("Order") you set the order in which the assays are placed side by side on the plate.

7. Entering samples

Manually / with a barcode scanner

- Click a well on the **sample plate** and type/scan.
- In the **field detail (sample)** on the right you can edit per well: **name, note, sample ID, prep notes**.
- **Return / Enter** jumps to the **next sample** (next well) – in the input direction. The cursor is then immediately in the name field, so you can scan/type right on.
- **Tab** switches between the fields within a well.
- **Prep notes** is a compact note field (one line high, longer notes scroll); there, **Shift+Enter** inserts a line break (a plain Enter jumps on).

Input direction (button at the top of the sample-plate column): by column (A1->B1->...) or by row (A1->A2->...).

Moving samples (drag & drop)

- Drag an **occupied well** with the mouse onto another: an **empty target** = the sample is *moved*, an **occupied target** = both samples are *swapped* (incl. note, sample ID, prep notes). Reversible with "**Undo**".
- A **short click** (without dragging) selects the well for editing as usual. *Note:* it is best to reorder **before** transferring to the PCR plate – a PCR layout that has already been created does not adjust retroactively.

Pasting several cells from Excel (Ctrl+V or right-click)

- In Excel, select and copy a **column or row of sample names**. In WellDone click the **start cell** and press **Ctrl+V** (or right-click -> *Paste*): a column/row is distributed continuously from there in the **input direction**; a **block** (several columns × rows) is inserted **position-true** (the selected cell = top-left corner).

- Only **sample names** are pasted; empty Excel cells are skipped, names that are too short/invalid are marked orange. Anything that extends beyond the 96-well plate is reported. (For note/sample ID/prep notes still use *Import*.)

Import from file (*Import*)

- Supports the **MagNA Pure 96** export as **XML** (recommended – contains the plate barcodes) or as **CSV** (also generic CSV with well / name / sample ID).
- On **XML** import, the **source plate ID** (barcode of the eluate plate) and the **plate label** (order) are automatically transferred into the fields on the right next to the sample plate.
- **Flags in plain text:** if a sample carries a flag (e.g. R03), the explanation from the export is appended directly in the **prep notes** – e.g. "... *Failed R03 (Reagent Expiration Rule overridden)*". The explanations come from the FlagDefinitions section of the XML file (official source); a code without a definition is left unchanged.
- **PDF (📄):** the "PDF" button creates a documentation sheet of the sample plate (96-well grid with names) including **all MP96 run details** (kit, protocol, sample/elution volume, **reagent/lot barcodes, flag explanations**) – for traceability/archiving.
- **Undo (↶):** undoes the last big action (import / Excel paste / clear) step by step.
- *Clear samples* resets the sample plate.

Creating an MP96 order (optional) (*MP96 Order*)

- Generates the **order for the MagNA Pure 96** from the entered samples – so you enter the samples only **once** and use them for the purification **and** the PCR setup.
- **Preparation (one time):** in the MP96 software, export the `TemplateData.xml` via *Utilities -> Tools -> Create Order Template*. In the dialog, read it in via "**Load TemplateData.xml**" (the path is remembered) – it provides the valid **kits, tests, volumes, plates and internal controls**.
- **Per run:** choose kit -> test/protocol -> volumes -> input/output plate and assign an **order name**. **Mandatory fields are marked with an asterisk** – the "**Create order...**" button only becomes active once all of them are filled (internal control, fill volume, source plate ID, comment, operator are optional). An **order XML** is created. The **order name** is used as the file name and may therefore not contain the characters \ / : * ? " < > | .
- **Gap-free layout (important):** the MP96 requires that the samples lie **gap-free from A1 (by column)**, and it always processes **whole columns of 8**. If there are gaps, WellDone **warns** on export and offers "**Compact**" (compacts the samples gap-free from A1 – reorders the **real sample plate**, reversible with "**Undo**"; also reachable via **right-click** on the sample plate). If the number of samples is **not a multiple of 8**, WellDone points this out – the MP96 tops up the rest with system fluid and processes it along.
- Read this XML into the MP96 software via *Utilities -> Query -> Import* (file type XML) -> the run is created. Unfilled wells are written with an empty SampleID (exactly as with the Excel template).
- **Presets:** save frequently used protocols as a **preset** – store the "**Preset**" field at the bottom of the dialog with *Save...* and load it later from the dropdown; then only samples + order name remain to be entered. *Delete* removes a preset.

Targeted selection (transfer only certain samples)

- With **Ctrl+click** select individual wells, with **Shift+click** a range (as in Excel).
- If a selection is active, **only** those samples are transferred into the plate setup. Without a selection, **all** samples are transferred.

8. Running the plate setup

The large blue button "**Run plate setup**" transfers your samples × selected assays onto the PCR plate.

Run model: per assay there is **one** positive control at the top (PTC/PPC), then the samples run through continuously, with **one** negative control at the end (NPC/NTC). Several assays are placed side by side in adjacent columns (or rows); the same sample sits at the same height per assay.

You set the **order of the controls** – or fixed well positions – per assay via **control placement** (section 6). With **fixed positions**, the samples flow around the control wells.

Start well: click an **empty** PCR well – the layout starts there (the hint right below the blue button shows the selected start well, e.g. "C5"). Without a click it starts at A1.

Layout (toggle): the button shows the current direction: **"by column"** fills column by column downward (A1->B1->C1 ...), **"by row"** fills row by row (A1->A2->A3 ...). One click switches it. When a column/row overflows, it continues at the top of the next one.

Replicates: every click on *Run plate setup* transfers the (selected) samples – including ones already placed. So **replicates/triplicates are possible at any time**. If you want to transfer only certain samples, mark them in the sample plate beforehand (click / Ctrl-click / Shift-click). Source wells already transferred to the PCR plate are **greyed out** in the sample plate – only for orientation; they can still be marked and transferred again (as a replicate).

Transferring / correcting multiple times:

- *Run plate setup* can be clicked several times – each new layout is appended to the next free column/row.
- **Controls stay unique per assay:** even across several steps there is **exactly one** positive control (at the front) and **one** negative control (at the back) per assay. With each further step **no** second PTC is set, and the NTC **moves** automatically to the new end.
- **Connecting gap-free:** every further step continues the layout **exactly where the NTC was before** – i.e. without a gap, like one continuous run. (If you deliberately want to continue elsewhere, click an **empty** PCR well as the start point beforehand.)
- *Undo* reverses the last layout, *Clear PCR* empties the whole PCR plate.
- **Deleting a single field:** click an occupied PCR field -> in the **field detail (PCR)** on the right click **Delete this field**. The field becomes free (sample names/MasterMix adjust), restorable with *Undo*. Not available in **384 quadrant mode** (there the samples lie in fixed 2x2 blocks).

Overflow: if not everything fits on the plate, the warning "*PCR plate full — not all samples fit. Please split the samples or choose an earlier start point.*" appears.

384-well plates

WellDone! can also lay out **384-well plates**.

- **Switching 96 384:** via the **tabs "96-well" / "384-well"** at the top of the assay list. This switches the PCR plate **and** the assays together (LCAPs for 96 and 384 are **different**, so every assay belongs to exactly one format). A switch **clears** the current PCR layout (with a confirmation).
- The **input/sample plate stays 96** – a 384 is filled from 96 source plates (see below). At 384 the plate shows 16x24 wells; for space reasons each cell shows only the **source coordinate** (e.g. H12) – the sample name on mouse-over/click and in the PDF as a **source-plate grid** (page 2).

Two layout modes (only at 384, toggle below the PCR plate):

- **"Quadrant: OFF" (continuous)** – like 96, just bigger: fills A1...P24 through (for hand pipetting). Here by-column/by-row is also selectable.
- **"Quadrant: ON" (default)** – for filling with a **96-channel pipette**. Here the samples **do not flow**: each source well corresponds fixedly to a **2x2 block** on the 384 plate (A1->{A1, A2, B2, B1}, H12->{O23, O24, P24, P23}). **Each active assay gets one corner of the block:**
 - clockwise: assay 1 -> A1 corner, assay 2 -> A2 corner, assay 3 -> B2 corner, assay 4 -> B1 corner,
 - counter-clockwise: A1 -> B1 -> B2 -> A2.

You set the direction with "**clockwise / counter-clockwise**". This way up to **4 assays** of the same sample lie close together. If fewer than 4 assays are active, a further transfer (e.g. a second source plate) fills the **next free** block corners. After 4 occupied corners the plate is full.

Controls in quadrant mode: for the assays concerned, choose the control placement "**Samples -> positive -> negative**" (or Neg->Pos). Then the samples are stamped 1:1 and the controls land in the **next free** block(s) after the samples (section 6).

More than 96 samples / several assays (batch workflow): load the first 96 source plate -> *Run plate setup* -> **clear** the sample plate -> load the next one -> run again. Each transfer fills the next free quadrant corners (or, in continuous mode, gap-free onward).

9. MasterMix calculation

For each active assay the required MasterMix volume is calculated:

$$\text{Reactions} = \text{number of samples} + \text{number of active controls} + \text{safety volume}$$

The **safety volume** is the reserve against pipetting loss. It is set **per assay** in the Edit dialog (at the very bottom under "reaction composition") – default **1.0**, possible values **0 to 10** in steps of 0.1 (e.g. 0.5). It affects **only the total volumes** (water, MasterMix ...), not the volume per single reaction.

- Example with PTC + NTC (2 controls) and safety volume 1.0: samples + 3
- Example with PTC + PPC + NPC + NTC (4 controls) and safety volume 1.0: samples + 5
- Safety volume 0 = no reserve (samples + controls); 0.5 = half a reaction extra.

Each component (water, primer/probes, MasterMix, RT enzyme) is multiplied by this reaction count. The overview "**Assays on PCR plate**" (right column) shows per assay: *N samples, K controls, (S standards,) X µl MasterMix* – the standards appear only if any are on the plate. All numbers with a decimal point in English mode.

10. Export: PDF, printing, plate-setup CSV

At the bottom right there are three buttons:

- **Print** – creates the pipetting-scheme PDF and opens it (for printing).
- **PDF** – saves the **pipetting scheme** as a PDF. It fits on **one A4 page** (landscape) and contains the plate header (name, plate ID, comment), the MasterMix table and the plate overview.
- **Setup PRO** – saves the **plate-setup CSV** for import into the LightCycler PRO software.

CSV format (columns): PlateId, PlatesetupName, PlateType, SampleId, WellPosition, AssayName, Targetname, Dilution, MasterMixName, LotNumber, RunProfileName, SampleRole.

- PlatesetupName = your **session name**.
- PlateType = "96" or "384" (depending on the chosen plate size); SampleRole = PTC/PPC/NPC/NTC/Unknown.
- **Sample IDs are truncated to 23 characters** (a requirement of the LightCycler PRO software): longer names are cut off **at the end** (the beginning is kept). WellDone! **points this out before the export** and names the affected samples – you can then cancel and shorten the names. Keep IDs as short and unique as possible.

Folder memory: every file button remembers **its own** last-used folder. So, for example, the plate-setup CSV can always land in the USB folder pLate_setup, the PDF in the print folder and the saved session somewhere else – on the next click each button suggests "its" location again.

11. At the LightCycler PRO: importing & starting the run

Once you have exported the **plate-setup CSV** (section 10), you continue at the LightCycler PRO software. This short guide summarizes the steps **from import to run start**. The Roche user assistance on the instrument is always authoritative.

Note on labels: the exact on-screen wording depends on the language set on the instrument. The English terms below are a guide; if your device runs in another language, the labels may differ.

Preparation

- Copy the CSV onto a **USB stick** into a folder named pLate_setup (it must be in the root directory of the stick – that is what the PRO software expects). Alternatively, transfer via SFTP.
- Make sure the matching **analysis package** is **installed and activated** on the PRO – otherwise the import or the run will fail.
- The CSV separator must be a **comma**. WellDone! already writes the file that way.

A • Import the plate profile

1. Open the **Plates** application.
2. Tab **Plate profile list** -> button **Import**.
3. Choose the storage location (USB or SFTP) and select the **CSV file**.
4. Choose **Import**. The plate profile then appears in the list.

B • Validate the plate profile

- **Validate** the imported plate profile in the instrument software (button **Validate**) – even if it has already been validated elsewhere. Missing controls can be added via the tab **Unassigned controls/standards**.

C • Load the plate

1. **Overview** application -> **Open loading drawer**.
2. Insert the sealed multiwell plate – **bevelled corner front left**.
3. **Close the loading drawer**. The instrument automatically reads the **barcode** and checks the format and the seal. If the plate ID matches a plate profile, it is **assigned automatically**.

D • Assign profiles (only if not automatic)

- In the **Run workflow** panel, assign the profile under **Plate profile** and the matching run profile under **Run profile**.

E • Start the run

- Choose **Start run**. The instrument status changes **Ready** -> **Warming up** (up to about 5 minutes) -> **In progress**. You can follow the progress in real time via **View run data**.

Prerequisites for the start: the thermocycler front flap closed, the plate loaded sealed, a run profile assigned. For the result calculation the run profile must contain **at least 15 cycles**.

Source of the instrument steps: *LightCycler PRO ISW, software version 1.2 / document version 2.0*. With a different software version individual labels may differ.

12. Saving & loading a session

- Top left: **New** · **Load** · **Save**.
 - A .wds file (JSON) is saved with plate ID, session name, comment, input direction, selected assays, all samples and metadata.
 - **New** clears the samples and the PCR plate, deselects all assays and resets the session name to today's date.
 - The **session name** is suggested automatically (date YYMMDD + short name of the assays); you can overwrite it.
 - The program remembers the **last-used folder** of the file dialogs.
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13. Data & backup

- **Assay library:** file assays.db in the folder %APPDATA%\WellDone (independent of the .exe location; type %APPDATA%\WellDone into the Explorer address bar to get there). It is automatically preserved when you replace the .exe and is extended with new fields when needed. To back it up, simply copy it.
 - **Sessions:** .wds files wherever you save them.
 - **Backup tip:** occasionally copy assays.db + your .wds files to a network drive/USB.
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14. Important limitations & conditions

- **Qualitative + quantitative (external standard curve):** qualitative tests (e.g. LightMix Modular from TIB Molbiol) and absolute quantification / MCGT with a standard curve (RS/MS) imported **on the instrument**. **No** in-run standard curves / dilution series in the plate setup.
 - **96- and 384-well plates** (LightCycler PRO). Switch via the assay tabs; the input/sample plate stays 96 (384 is filled from 96 source plates, see section 8).
 - Only **assays with the same PCR profile** at the same time.
 - At most as many wells as the **chosen plate** holds (96 or 384), including all controls; on overflow a warning appears (no automatic splitting across several plates).
 - **Sample ID / sample name:** at least **2** and at most **23 characters** and **none** of the special characters ^ ~ \ & | " , ; (requirements of the LightCycler PRO software). Names that are too short or invalid are marked **orange** in the sample plate and reported before the layout. **Plate ID** and **plate name** are also checked (their own, slightly different character sets) and reported before the CSV export.
 - **Display name** of an assay limited to 24 characters.
 - **Language: German (default) or English**, switchable in the "" **dialog** (at the top, *Sprache / Language*); the change takes effect after a **restart** and affects the interface, the PDF and all messages. Numbers appear per language with a **comma** (German) or a **point** (English). The manual is available in German and English and opens automatically in the selected language.
 - WellDone! is **not validated** – see section 2.
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15. Troubleshooting (FAQ)

- **"I can't tick an assay."** -> It has a different PCR profile than the ones already active. Deselect the others first.

- **"My sample is not transferred."** -> Is a **selection** (Ctrl/Shift-click) perhaps active that excludes it? Cancel the selection by clicking an empty well. (Replicates are generally allowed – samples already placed are no longer blocked.)
 - **"The layout starts in the wrong place."** -> Before running, click an **empty** PCR well as the start point.
 - **"Print only opens the PDF."** -> That is intended; in the PDF viewer then go to Print.
 - **"The PDF opens as unreadable gibberish (or in the editor/Notepad)."** -> No real PDF viewer is set as the default program for .pdf on the computer. In Windows under *Settings -> Apps -> Default apps -> Default by file type* -> .pdf choose a PDF viewer (e.g. **Microsoft Edge** or **Adobe Reader**). The pipetting scheme itself is correct — it is only displayed incorrectly.
 - **"Saving PDF/CSV seems to do nothing."** -> Choose a folder you are allowed to write to (e.g. **Documents** or **Desktop**) — not "Program Files" or a write-protected drive. If the file is still open in another program, close it first.
 - **"assays.db gone / library empty."** -> The library is **not** next to the .exe but under %APPDATA%\WellDone!\assays.db (type %APPDATA%\WellDone into the Explorer address bar) – independent of the .exe location. On a new computer a starter set is created automatically on the first start.
 - **"Sample ID is truncated."** -> A limit of the LightCycler software (23 characters); WellDone! truncates **at the end** (the beginning is kept) and points out the affected samples before the CSV export. Choose shorter, unique IDs.
 - **"A sample has an orange outline."** -> The sample name has only 1 character. The LightCycler PRO needs **at least 2 characters** per sample – extend the name accordingly (a note also appears before the layout).
 - **"There is a gap / an empty field on the plate."** -> Normally further steps connect gap-free (the layout continues where the NTC was before). A gap arises only if you deliberately click an empty well further back as the start point – that is allowed.
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16. License & distribution

- You **may pass** WellDone! on to colleagues and use it in the lab. Private/free distribution is unproblematic.
- WellDone! uses open-source libraries (not a proprietary part of Roche):
- **PySide6/Qt** (© The Qt Company) — license **GNU LGPL v3** · <<https://www.qt.io/licensing>>
- **ReportLab** (PDF generation) — **BSD license** · <<https://www.reportlab.com>>

These licenses permit use in WellDone! without disclosing your own source code. As long as the app is passed on **as an unchanged, standalone** .exe, all license obligations are met. (You will find the same note in the **"Help / About"** dialog of the app.)

- **Roche** and **LightCycler** are trademarks of their respective owners. WellDone! is an **independent, private project**, not reviewed or endorsed by Roche.
 - **Sale / commercial distribution** is not permitted!
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17. Feedback & contact

Suggestions for improvement and bug reports are very welcome:

- **E-mail:** well.done.pcr.setup@gmail.com
- **Online form:** forms.gle/9QYvGAW1AXsfxYRk8

In WellDone! you can reach these contacts at any time via the **"Help / About"** button at the top of the plate-setup panel.

18. Version history

Version 10.4 (June 2026)

- **English manual (new):** the manual is now also available in **English** and opens automatically in the selected language (German interface -> German manual, English interface -> English manual).
- **Consistent colors in the pipetting scheme:** the **MasterMix table** now uses the same colors as the **plate layout** – the "total" column in the **full assay color** of the sample wells, the "per 1 rxn" column in the **lightened NTC shade** (text automatically light/dark for good readability). This lets you match the table and the plate at a glance.
- **Renamed:** the sample-plate button is now "**MP96 Order**" (was "Export"), and the action button in the dialog is "**Create order...**" (was "Export...").
- **MP96 order from WellDone (new):** WellDone now also creates the **order for the MagNA Pure 96** – button "**MP96 Order**" at the sample plate (row *PDF · Import · MP96 Order*). A loaded *TemplateData.xml* (from the MP96 software) provides the valid **kits, tests, volumes, plates and internal controls**; the samples come automatically from the sample plate. **Mandatory fields are marked with an asterisk** (export is only possible once all are filled). Export as an **order XML** that the MP96 reads via *Utilities -> Query -> Import* – so from **one** sample entry you get, in one go, the MP96 order **and** (as before) the pipetting scheme + LC-PRO setup.
- **Presets:** frequently used protocols (kit/test/volumes/plates/IC) can be **saved and loaded as a preset** ("Preset" field at the bottom of the dialog) – per run only samples + order name remain to be entered.
- **Tidied up:** the former "**Template**" button (CSV template of the sample plate) has been removed; the sample-plate buttons are now *PDF · Import · MP96 Order / Undo · Clear samples*.
- **MP96 layout check:** on export WellDone checks whether the samples lie **gap-free from A1 (by column)** (required by the MP96) and whether the number is a **multiple of 8**; otherwise there is a warning. New: "**Compact**" moves the samples together gap-free from A1 (reorders the real sample plate, reversible with "Undo") – from the warning **or** via **right-click** on the sample plate. The **order name** is checked for characters that are illegal in file names.

Version 10.3 (June 2026)

- **English language variant:** WellDone! now speaks **German and English** — switchable in the **"" dialog** (at the top, *Sprache / Language*, default German). The change affects the interface, the pipetting-scheme PDF and all messages and takes effect after a **restart**; in English mode numbers are written with a **point** instead of a comma (also in the input fields). **Assay, sample and instrument data** (names, LCAP, PCR profile, Roche CSV columns) remain unchanged in both languages. The manual is for now German only.
- **Reorder samples by drag & drop:** in the sample plate an occupied sample can be dragged onto another well with the mouse — **empty target = move, occupied target = swap** (incl. note/sample ID/prep notes). Reversible with "**Undo**". A short click edits the well as usual; the PCR plate stays unchanged (read-only).

Version 10.2 (June 2026)

- **Central assay library (network template):** new button "**Template**" in the assay column (second-to-last row: *Template · Duplicate*). With it the assay library can be **loaded from a template on a network share** (completely replaces the local library — **a dated backup** *assays.backup-...db* **is created automatically beforehand**) or the local library can be **saved as a template** (publish a master for all computers). The local *assays.db* remains the working copy (always runs, even offline); a **fresh computer** without a local library is filled automatically from the template on the first start. Synchronization is manual only — no shared live file, no write/lock conflicts.

Version 10.1 (June 2026)

- **MP96 XML import:** the MagNA Pure 96 export can now be read as XML – incl. **source plate ID** (eluate barcode) and **plate label** (order), which are transferred automatically into the new fields on the right next to the sample plate. CSV remains; TXT is dropped (= CSV). **Flags in the prep notes are translated to plain text** – the explanation from the FlagDefinitions section of the XML is appended after the code (e.g. "R03 (Reagent Expiration Rule overridden)"); unknown codes are left unchanged.
- **PDF:** page 2 now shows the source plate as a **96-well grid with sample names** (instead of a table); the plate label appears in the header and heading. **Standards legend** multi-column (max. 3 lines -> stays on page 1) with a **color dot** per assay. New button "**PDF**" creates a sample-plate documentation sheet (96-well grid + all run details incl. **reagent/lot barcodes and flag explanations**, for traceability).
- **Sample-plate tools tidied up:** rows "**Template · PDF · Import**" (each) and "**Undo · Clear samples**" ($\frac{1}{2}$ each); a new "**Undo**" reverses big actions (import / paste / clear); "Plate details" is dropped (it is all in the PDF). Excel paste now also works via **right-click -> Paste** (was not recognized in German). **Prep notes shrunk to one line** -> the filled field detail is no higher than the empty one, and the blue "**Run plate setup**" button no longer jumps down when you click a sample field (the PCR menu stays on the midline). Plate-ID hint as a placeholder; comment field one line higher.

Version 10.0 (June 2026)

- **Manual corrected:** outdated notes "96-well only / 384 not supported" removed – **384-well has been supported since 9.3**; the well limit is now format-dependent (96 or 384). Scope & limitations adapted to the **external standard curve** (since 9.4): qualitative **and** quantitative tests (RS/MS, standard curve on the instrument) – **no** in-run standard curves/dilution series in the plate setup.
- **Bug fixed (sample plate):** Ctrl/Shift multi-selection deleted previously marked samples (the hidden inline editor overwrote them with empty text) – fixed. Source wells already transferred to the PCR plate are **greyed out** again.
- **New:** individual occupied **PCR fields can be deleted** – click the field -> in the field detail ; restorable with "Undo"; not available in 384 quadrant mode.
- **New:** **paste several cells from Excel** – click the start cell and Ctrl+V; a column/row is distributed in the input direction, a block position-true (sample names only).

Version 9.4 (June 2026)

- **Standards (external standard curve):** per assay, optional reference/melting standards (RS/MS) with a target name + an optional note (documentation); they occupy a well and appear in the CSV with SampleRole RS/MS and Targetname. For abs. quantification / MCGT with a standard curve imported on the instrument. (Qualitative + Tm calling need no standards.)
- **Assay management:** "Duplicate" button (duplicate an assay) and **double-click** to edit.
- Standards are colored **darker** like positive controls; the overview "Assays on PCR plate" and the **standards legend** in the PDF (left-aligned, multi-column when many) now list them too. **Bug fixed:** in quadrant mode controls/standards are now always placed (even if a sample sits in the last well H12). PDF **sample assignment** always on page 2, left-aligned, up to 3 columns, with the hint "sorted by column" (without zebra).

Version 9.3 (June 2026)

- **384-well layout** reworked: switching 96/384 via the **assay tabs**; **quadrant mode** stamps each source well into its 2x2 block (96-channel pipette), up to 4 assays per block, direction clockwise/counter-clockwise.
- New control placements "**Samples -> positive -> negative**" and "**Samples -> negative -> positive**"; fixed control wells now as **text input**.
- **384 grid** shows the **source coordinate**; PDF with a "**Sample assignment**" section and footer "Page X of Y".
- **Duplicates toggle removed** – replicates are always allowed. Comment field enlarged.

Version 9.2 (June 2026)

- This version overview added to the manual.

Version 9.1

- **Control placement per assay** selectable: positive->samples->negative, reversed, or **fixed positions** (freely chosen wells via a mini plate).
- **Safety volume per assay** adjustable (reserve against pipetting loss).
- File dialogs remember the last-used folder **per purpose**.
- Sample-plate buttons unified (Import / Clear).

Version 9.0

- Alignment with the official **Roche CSV specification**; check for **illegal characters** in sample/plate names (marking + warning).

Version 8.9

- Sample names **at least 2 characters** (the LightCycler PRO does not accept 1-character IDs); names that are too short are marked orange.
- Plate-ID help text; note "qualitative tests only"; tooltips in the assay dialog.

Version 8.8

- Controls **only once** per assay — even across several layouts (exactly 1 PTC + 1 NTC).
- Note before the export when a sample name longer than 23 characters is truncated.

Version 8.7

- Positive/negative controls **clearly marked** on screen and in the PDF (symbol + shading).

Version 8.6

- Automatic **update check** (note in the "" dialog); distribution via the shared folder.
- More robust PDF/CSV creation (clear error messages, storage location "Documents").
- Per-user assay library (%APPDATA%).